

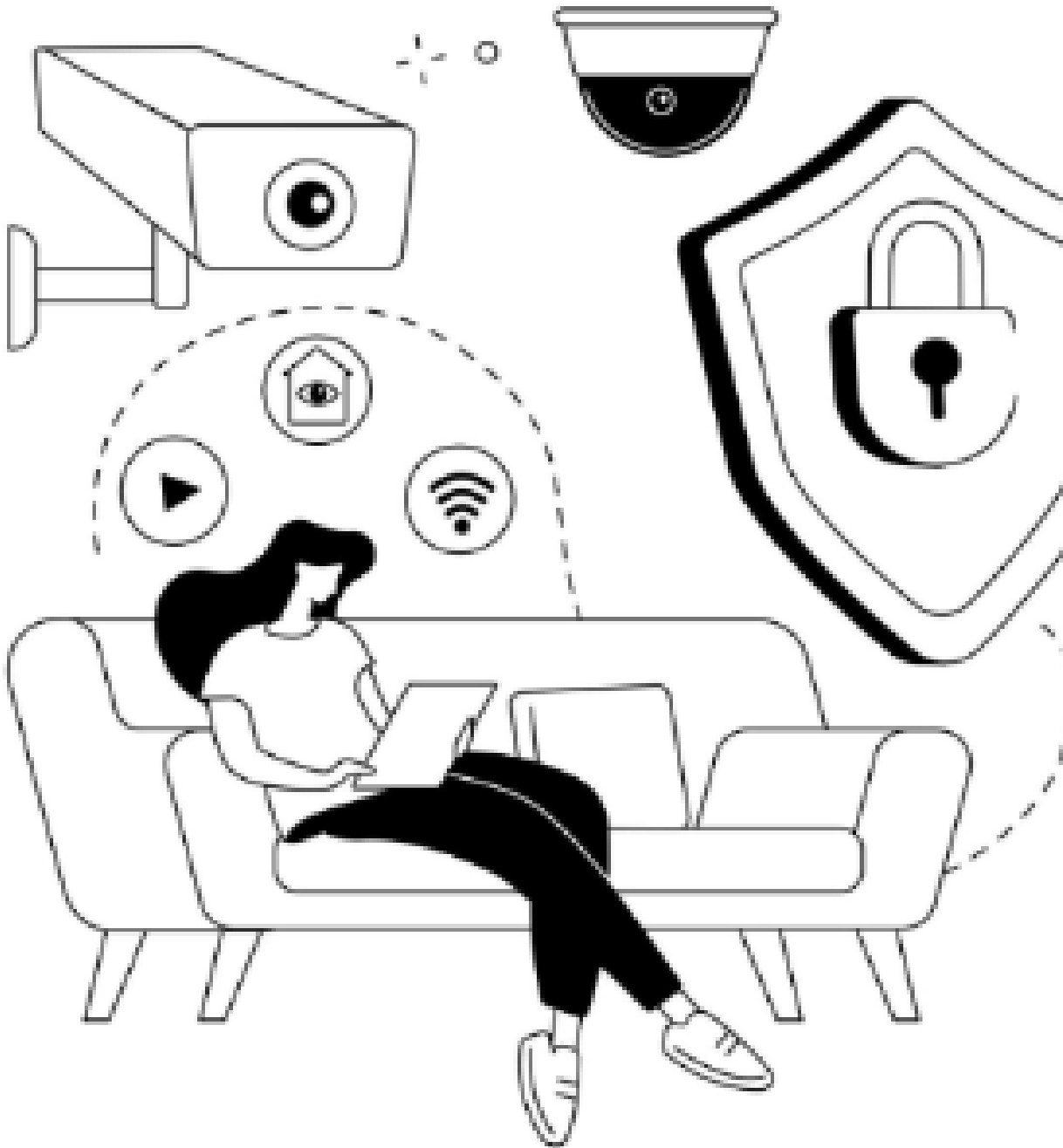
Smart Security System

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Project Overview

The Smart Security System is an IoT-based solution designed to enhance security and safety in indoor environments.

The system uses an ESP32 microcontroller connected to multiple sensors and actuators to:

- Detect unauthorized motion using a PIR sensor.
- Trigger local alarms using a buzzer.
- Monitor ambient temperature.
- Allow secure access using a keypad.
- Send real-time sensor data to the FIWARE Orion Context Broker.
- Provide remote monitoring through cloud-based dashboards.

This project demonstrates IoT concepts including sensing, communication, cloud integration, real-time monitoring, and event-driven automation.

Problem Statement

Traditional security systems are often expensive and lack flexibility.

Challenges:

- Intrusion detection requires continuous monitoring.
- Temperature anomalies may indicate fire hazards.
- Unauthorized access should trigger alerts immediately.
- Users need real-time visibility into security events.
- Our solution addresses these issues through a low-cost IoT architecture based on ESP32 and FIWARE.

Project Objectives

Security Objectives

- Detect human motion in monitored areas.
- Trigger alarms upon intrusion.
- Authenticate users through a keypad.

Monitoring Objectives

- Continuously monitor room temperature.
- Upload sensor readings to the cloud.

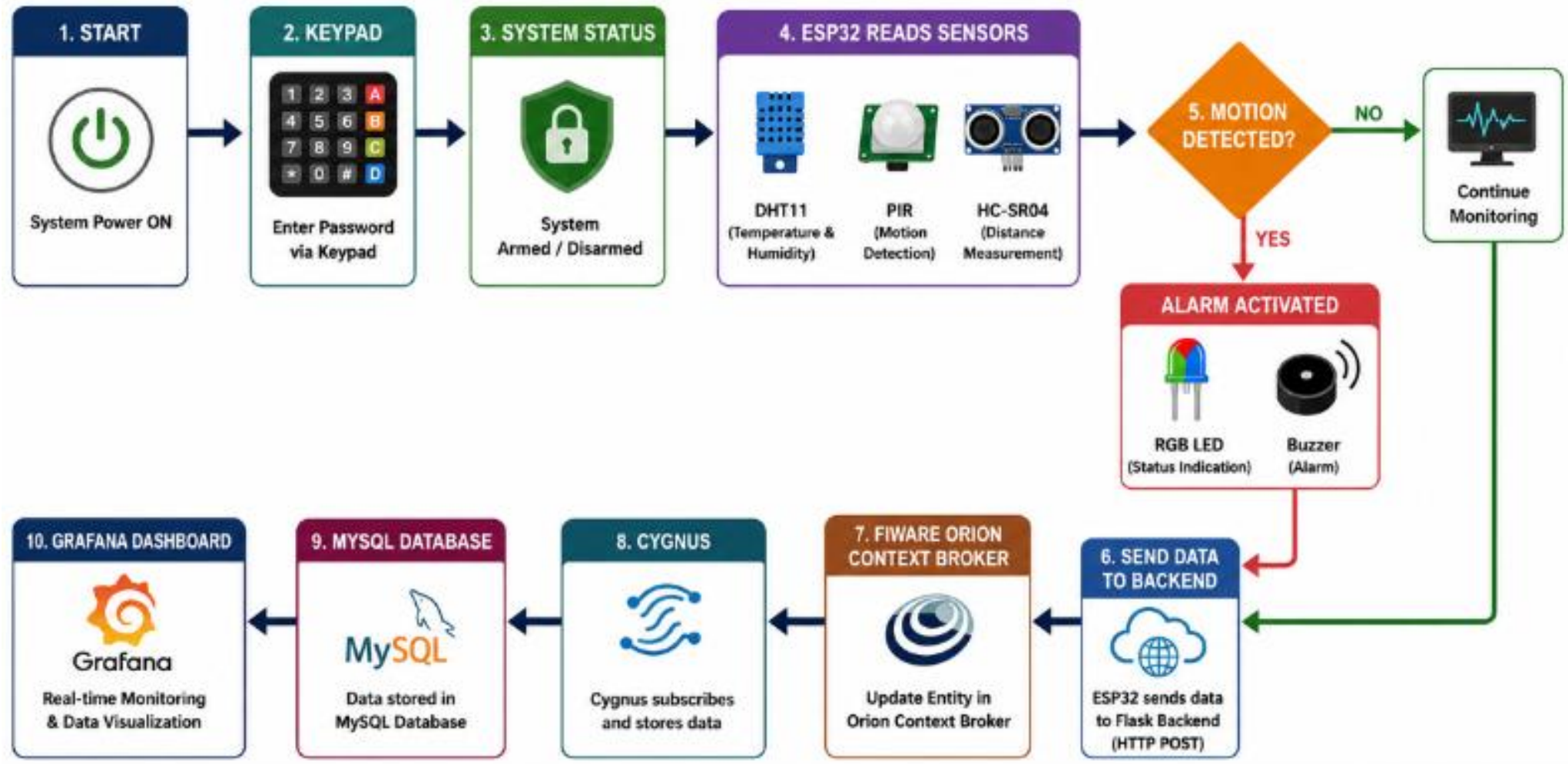
IoT Objectives

- Connect devices to Wi-Fi.
- Integrate with FIWARE Orion Context Broker.
- Visualize data remotely.
- Demonstrate real-time IoT communication.

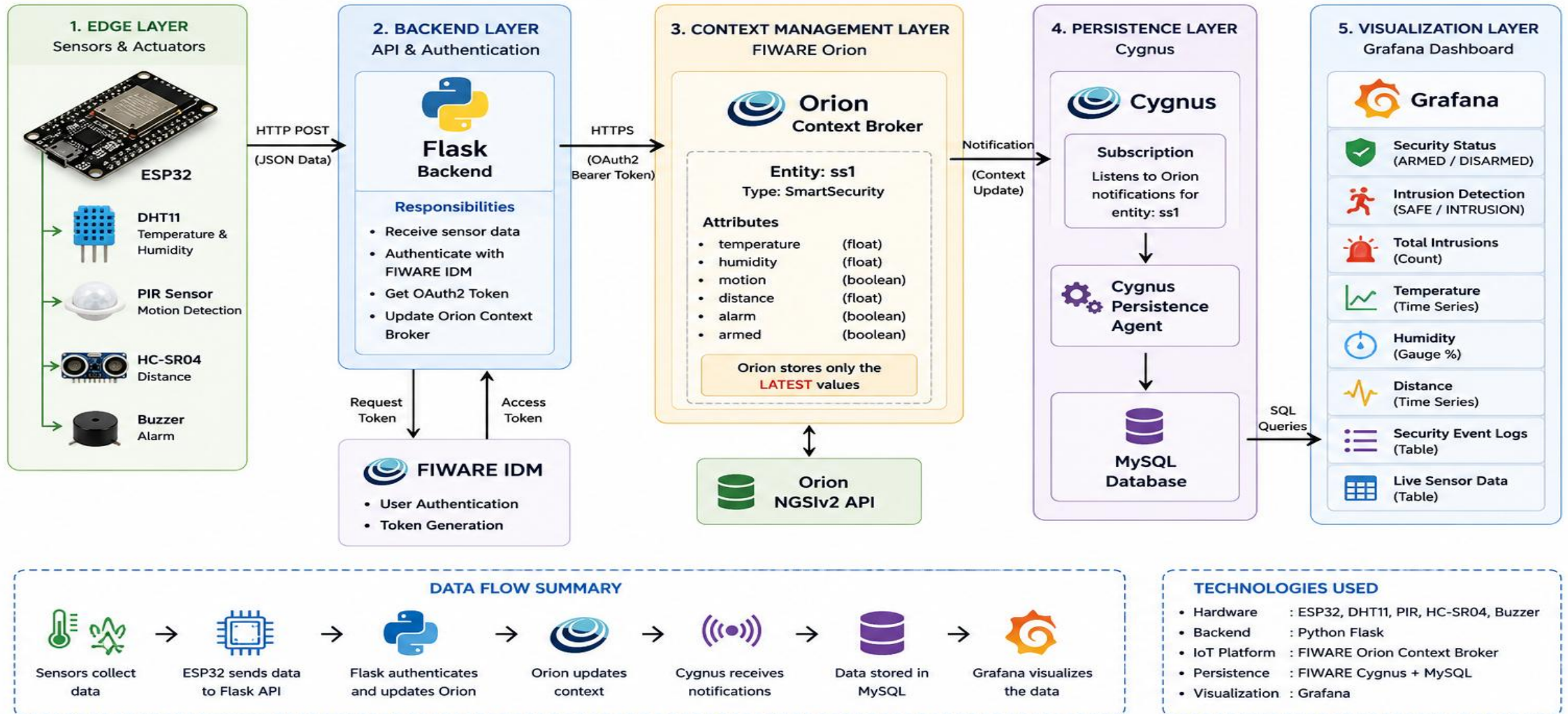
Use Cases

- Smart Home Security
- Office Security
- Laboratory & Server Room Monitoring
- Warehouse Monitoring
- Healthcare & Pharmacy Storage
- Educational Institutions
- Remote IoT Monitoring

System Workflow



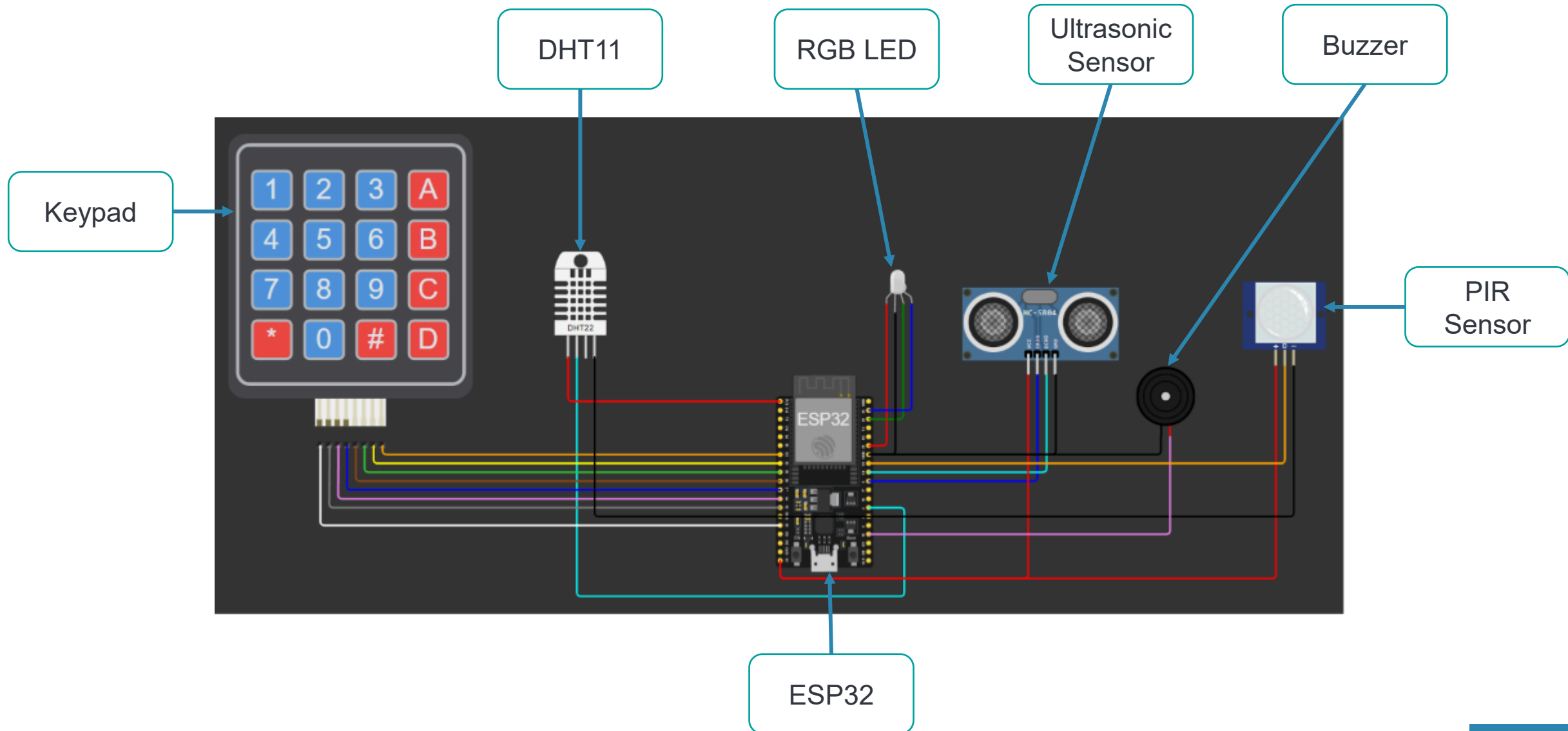
System Architecture



Hardware Components

Hardware Component	Purpose
ESP32 DevKit	Main controller for sensor integration and communication
DHT11 Sensor	Measures temperature and humidity
PIR Motion Sensor	Detects human movement
HC-SR04 Ultrasonic Sensor	Measures object distance
4×4 Keypad	Arms and disarms the security system
RGB LED	Indicates system status (armed, disarmed, alarm)
Active Buzzer	Generates audible alarm notifications
Breadboard	Prototype circuit assembly without soldering
Jumper Wires	Electrical connections between components

Fizzing Diagram



Implemented Features

Real-Time Monitoring

-  Temperature
-  Humidity
-  Motion
-  Distance

Smart Security

-  Arm/Disarm
-  Buzzer Alarm
-  RGB Status

Data Management

-  HTTP Communication
-  MySQL Storage
-  Dashboard

PIR Motion Detection

Working Principle

- PIR sensor detects infrared radiation changes.
- Human movement changes the IR pattern.
- ESP32 receives HIGH signal.
- Motion event is generated.

Actions

- Motion detected message.
- Buzzer activated.
- Event uploaded to FIWARE.

Intrusion Alert using Keypad

Feature Added

The keypad introduces an authentication mechanism.

Working

- User enters PIN.
- ESP32 verifies PIN.
- If PIN is correct:
 - Access granted.
 - Alarm disabled.
- If PIN is incorrect:
 - Intrusion alert generated.
 - Buzzer sounds.
 - Event uploaded to FIWARE.

Benefits

- Prevents unauthorized access.
- Adds another security layer.

Temperature Monitoring

Feature Added

- Temperature sensor continuously measures ambient temperature.

Functions

- Read temperature every few seconds.
- Display readings on Serial Monitor.
- Send temperature values to FIWARE.

Safety Scenario

If temperature exceeds threshold:

- Warning generated.
- Potential fire hazard indication.
- Alert can be sent to dashboard.

FIWARE Integration

Why FIWARE?

FIWARE provides:

- Context management
- IoT data handling
- Real-time updates
- Open-source ecosystem

Orion Context Broker

- Stores entities such as:

```
{  
  "id": "urn:ngsi-Id:MotionSensor:001",  
  "type": "MotionSensor",  
  "motion": {  
    "value": true  
  }  
}
```

Data Persistence

Sensor readings are automatically stored in MySQL using Cygnus.

Stored Information:

- Timestamp
- Temperature
- Humidity
- Distance
- Motion Status
- Armed Status

This enables historical analysis and long-term monitoring.

Grafana Dashboard



Dashboard Panels

- **System Status**

Displays: SYSTEM ARMED

Indicates whether the security system is active.

- **Intrusion Detection**

Shows the latest motion detection status.

Values: SECURE or INTRUSION DETECTED

- **Total Intrusion Count**

Displays the total number of times motion was detected.

Purpose:

- Security analytics
- Event frequency monitoring

Dashboard Panels

- **Humidity Monitoring**

Visualization Type: Gauge

Thresholds:

Range	Status
0 – 30 %	Dry
30 – 60 %	Normal
60 – 100 %	Humid

- **Distance Monitoring**

Visualization Type: Time Series Graph

Displays object proximity measurements from the ultrasonic sensor.

- **Security Event Log**

Contains:

- Timestamp
- Sensor Attribute
- Sensor Value

Provides historical records of all security events.

Results

Achievements

- Real-time monitoring
- Motion detection
- Data persistence
- Dashboard visualization
- Historical analysis

Benefits

- Scalable
- Modular
- Low-cost
- Easy to extend
- Cloud-ready

Future Enhancements

- Camera integration.
- Mobile notification system.
- Face recognition.
- RFID authentication.
- SMS/Email alerts.
- Cloud analytics dashboard.
- Mobile application.

Conclusion

This project successfully demonstrates a complete IoT-based security and monitoring solution using ESP32 and FIWARE.

The system combines:

- Motion detection
- Intrusion prevention
- Temperature monitoring
- Cloud connectivity

Through real-time sensing, cloud integration, and event-driven responses, the project showcases practical IoT concepts applicable to smart homes, offices, laboratories, and industrial environments.

The addition of keypad authentication and temperature monitoring further improves both security and safety, making the system more robust and scalable for future developments.

THANK YOU !!